How fast a wave travels

A wave travels one wavelength during each cycle.

Its speed is the frequency multiplied by the wavelength:

v = f times wavelength.

Frequency in hertz counts cycles per second. With wavelength in metres, the resulting speed is measured in metres per second.

For example, 5 Hz and 2 m give a speed of 10 m/s.

Time per cycle

The period is the time taken for one complete cycle.

Period and frequency are reciprocals: $T = 1/f$.

Doubling the frequency halves the period. For example, a 10 Hz sound has a period of 0.1 s and a 20 Hz sound has a period of 0.05 s.